Globalisation and Unequalisation: What Can Be Learned from Value Chain Analysis?

RAPHAEL KAPLINSKY

Although many have gained from the process of globalisation, there remains a stubbornly large number of people living in absolute poverty and a rise in inequality within and between countries. The issue is thus not whether to participate in the global economy but how to do so in a manner which provides for sustainable and equitable income growth. This study shows how value chain analysis can be used both to chart the growing disjuncture between global economic activity and global income distribution and to provide causal explanations for this outcome. In so doing, value chain analysis provides valuable insights into policy formulation and implementation.

I. INTRODUCTION

For many of the world's population, the growing integration of the global economy has provided the opportunity for substantial income growth. This is reflected not only in higher incomes, but also in the improved availability of better quality and increasingly differentiated final products. However, at the same time, globalisation has had its dark side. There has been an increasing tendency towards growing unequalisation within and between countries and a stubborn incidence in the absolute levels of poverty, not just in poor countries. These positive and negative attributes of globalisation have been experienced at a number of different levels – the individual, the household, the firm, the town, the region, the sector and the nation. The distributional pattern emerging in recent decades of globalisation is thus simultaneously heterogeneous and complex.

Raphael Kaplinsky, Institute of Development Studies, University of Sussex, Brighton BN1 9RE and Centre for Research in Innovation Management, University of Brighton, Sussex BN1 9PH. The author is grateful to John Humphrey, Hubert Schmitz, Hans Singer, Adrian Wood and an anonymous referee for their helpful comments on an earlier draft. He regrets that the remaining errors and weaknesses are his alone.

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If those who had lost from globalisation had been confined to the non-participants, the policy implications would be clear – take every step to be an active participant in global production and trade. However, the challenge is much more daunting than this, since the losers include many of those who have participated actively in the process of global integration. Hence, there is a need to manage the mode of insertion into the global economy, to ensure that incomes are not reduced or further polarised.

Three central questions arise from these observations. First, why has there been so little correspondence between the geographical spread of economic activity and the spreading of the gains from participating in global product markets? Secondly, to what extent is it possible to identify a causal link between globalisation and inequality? And, thirdly, what can be done to arrest the unequalising tendencies of globalisation? These three related questions have important methodological implications – what is the best way to generate the information required to document these developments in production and appropriation, and how can we identify policy instruments which might halt, and perhaps partially reverse these developments?

It is the central contention of this study that value chain analysis provides an important framework for addressing these crucial questions. In developing this argument, in section II we will posit a causal relationship between increasing inequality and the global integration of production and trade. In section III we outline the central elements of value chain analysis, show how it differs from conventional industry analysis, and suggest ways in which it augments the analysis of economics and other social sciences. This is followed in section IV by two brief short case studies which illustrate the contribution to be made by value chain analysis in understanding this causal relationship. The study concludes by drawing conclusions for both future research and policy design and implementation.

II. GLOBALISATION AND UNEQUALISATION

Globalisation is not a new phenomenon; it has ebbed and flowed over the past century. There are many measures of 'globalisation', none of which is free from imperfections. However, one indication of growing integration is the proportion of production which is traded; this has grown significantly in most parts of the global economy. At the same time, many indicators of inequality, within and between countries, have increased, and the absolute numbers of those living below $1 per day in 1985 purchasing parity prices have remained stubbornly large (at around 1.2 billion).[1]

One possible explanation for these rising levels of inequality is that globalisation has bypassed much of the world's population, particularly

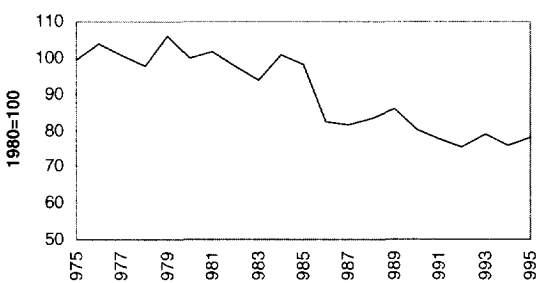
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those living in poor countries. Perhaps, but more and more economic activity in virtually all countries is affected by the global interchange of goods and services (in addition to other components of globalisation). Moreover, many of those countries that have suffered in distributional terms have seen a substantial increase in their trade/GDP ratios. For example, the trade/GDP ratio for sub-Saharan Africa rose from 51 to 56.1 per cent between 1985 and 1995, whilst its share of global output fell markedly [*Kaplinsky, 2000a*]. So, the explanation for increased inequality will have to also explain how it is that countries and regions can increase their participation in global exchange and at the same time see a decline in their relative income shares (and also sometimes even in their absolute standards of living). For this reason, the focus of attention must also lie with the mode in which firms, countries and regions participate in the process of global production and exchange.

An explanation for these declining country shares in global income in the context of growing participation in global markets can be found in the concentration of developing countries in commodity sectors. These have experienced declining terms of trade over a sustained period [*Singer, 1950; Prebisch, 1950*]. This is indeed a problem still encountered in many developing countries, particularly in SSA. As a consequence, it has long been held that structural change should aim wherever possible to encompass the transition from the growing and extraction of primary commodities to the manufacture of industrial products. But, here, since the mid-1980s we have seen an emerging trend for the terms of trade of a range of manufactured commodities – particularly those produced by developing countries – to decline (Figure 1).[2] It is significant that this decline in the terms of trade of developing country manufacturing exports coincides with China's entry into global markets.[3]

FIGURE 1

PRICE OF DEVELOPING MANUFACTURED EXPORTS RELATIVE TO DEVELOPED COUNTRY MANUFACTURED EXPORTS OF MACHINERY, TRANSPORT EQUIPMENT AND SERVICES



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TABLE 1

INCREASING COMPETITION AND DECLINING UNIT PRICES: THE CASE OF JEANS MANUFACTURING IN THE DOMINICAN REPUBLIC

	Volume (per week)	Unit price ($)
January 1990	9,000	2.18
October 1990	5,000	2.05
December 1990	3,000	1.87
February 1991	Arrangement terminated and assembly transferred to Honduras	
Total investment in equipment by Dominican Republic firm was US$150,000		

Source: Kaplinsky [*1993*].

The problem of falling returns not only confronts economies, but also individual firms. When firms confine their competences to the simple assembly of imported materials, they become subject to increasing competition and hence to falling returns. For example, in the Dominican Republic in the early 1990s, the assembly of jeans in export processing zones occurred in the context of intense regional competition (often surfacing in competitive devaluations), resulting in sustained falls in unit prices (Table 1).[4]

The consequence of the failure of individual firms, groups of firms and national economies to insert themselves appropriately into global markets is that the spectre is raised of *'immiserising growth'*. This describes a situation where there is increasing economic activity (more output and more employment) but falling economic returns. For example, over the past two decades Brazilian shoe producers have commanded more than 12 per cent of global leather shoe exports. At the same time, between 1970 and 1980 average real wages in the sector were stagnant, and during the following decade they fell by approximately 40 per cent in real terms [*Schmitz, 1995*]. In the Dominican Republic, real wages (as measured in international purchasing power) fell by 45 per cent during the second half of the 1990s, largely as a consequence of competitive devaluations in the region [*Kaplinsky, 1993*].[5]

Is it possible to determine those factors which drive the distribution of the gains from global production and exchange, explaining both why some parties have gained and others have lost from globalisation? Moreover, can we then use this analysis to identify policy levers – relevant at the level of individuals, households, firms, regions and countries – which may lead to a different, and more favourable distributional outcome? The objective of this study is to show that value chain analysis has an important role to play in meeting these objectives.

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III. THE CONTRIBUTION OF VALUE CHAIN ANALYSIS

Many factors associated with globalisation will affect the distribution of returns. For example, macro economic disturbances associated with capital mobility (and particularly capital volatility) can have major consequences for the living standards of many millions of people, as was the case following the Asian crisis of 1997 [*Griffiths-Jones and Cailloux, 1999*]. But, as we shall see in the case studies in section IV below, insofar as distribution is an outcome of the globalisation of production and exchange, value chain analysis provides a valuable methodological tool for explaining these developments.

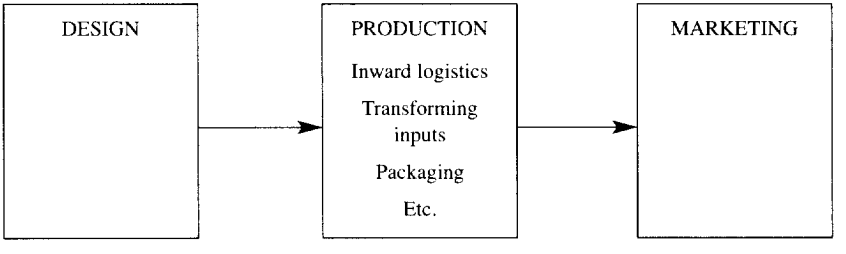
What is a Value Chain?

The value chain describes the full range of activities which are required to bring a product or service from conception, through the intermediary phases of production (involving a combination of physical transformation and the input of various producer services), delivery to final consumers, and final disposal after use. Considered in its most elementary form, it takes the shape as described in Figure 2, although in reality value chains are considerably more extended than this. As can be seen from this, production *per se* is only one of a number of value added links. Moreover, there is a range of activities within each link of the chain (only those for production are detailed in the Figure).

The concept of the value chain was used in the 1960s and the 1970s by analysts charting a path of development for mineral-exporting economies [*Girvan, 1987*]. It was also adopted in recent French planning literature in the form of the *filière*.[6] But during the 1990s, value chain analysis has become widely used, particularly as a consequence of the writings of Michael Porter [*Porter, 1985, 1990*] and in an influential book by Womack and Jones (who refer to it as the 'value stream') [*Womack and Jones, 1996*]. A further source of the recent prominence of the concept of the value chain,

FIGURE 2

A SIMPLE VALUE CHAIN



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particularly in relation to developing countries, arises from the work of Gereffi, building on world system analysis. We shall consider this literature in more detail below.

Considered in this way, the value chain is merely a descriptive construct, at most providing a heuristic framework for the generation of data. However, recent developments of the value chain framework have begun to provide an analytical structure which, as we shall see below, provides important insights into our twin concerns with the determinants of global income distribution and the identification of effective policy levers to ameliorate trends towards unequalisation.[7] There are three important components of value chains which need to be recognised and which transform an heuristic device into an analytical tool:

- value chains are repositories for rent, and these rents are dynamic;
- effectively functioning value chains involve some degree of 'governance'
- effective value chains arise from systemic, as opposed to point, efficiency.

Three Key Elements of Value Chain Analysis

Barriers to entry and rent:[8] The theory of economic rent was first formulated by Ricardo, who distinguished between rent as a factor income – '[i]n popular language, the term is applied to whatever is annually paid by a farmer to his landlord' – and economic rent – 'Rent is that portion of the produce of the earth which is paid to the landlord for the use of the original and indestructible powers of the soil' [*Ricardo, 1817: 33*]. Here Ricardo was highlighting the significant role played by scarcity, since economic rent does not arise from the differential fertility of land itself (which was central to Ricardo's analysis), but from unequal access to this resource.

But as Schumpeter showed, scarcity can be constructed. That is, it can arise from purposive actions rather than as a consequence of the bounty of nature. For Schumpeter, the entrepreneur played a unique role in 'the carrying out of new combinations' [*Schumpeter, 1961: 107*]. Entrepreneurial surplus is the return to the innovation of a 'new combination' and arises when the price of the product following the introduction of the 'new combination' provides greater returns than are required to meet the cost of the innovation. These returns to innovation are a form of super-profit and act as an inducement to replication by other entrepreneurs:

Thus, in summary:

- economic rent arises in the case of differential productivity of factors (including entrepreneurship) *and* barriers to entry (that is, scarcity);

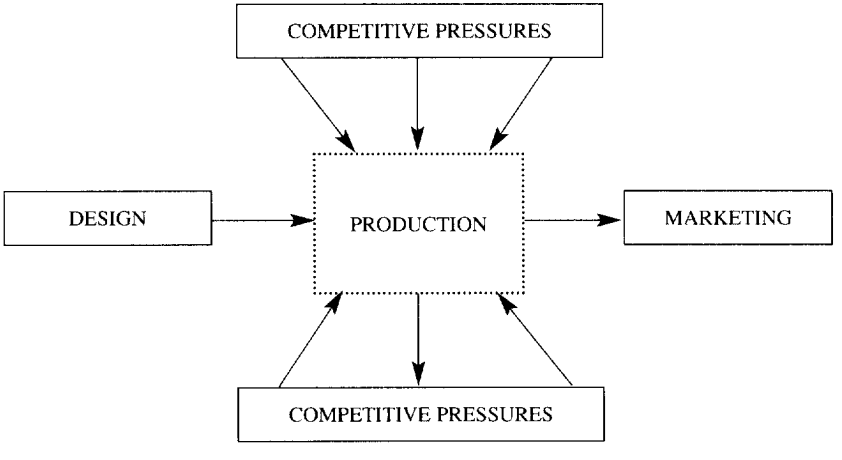
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- economic rent may arise not just from natural bounty, but also as producer surpluses that are created by purposive action.[9] These augmented rents have become increasingly important since the rise of technological intensity in the mid-nineteenth century [*Freeman, 1976*] and the growth of differentiated products after the 1970s [*Piore and Sabel, 1984*];
- most economic rent is dynamic in nature, eroded by the forces of competition. Producer rent is then transferred into consumer surplus through the process of competition;
- the process of competition – the search for 'new combinations' to allow entrepreneurs to escape the tyranny of the normal rate of profit, and the subsequent bidding away of this economic rent by competitors – fuels the innovation process which drives capitalism forward.

As more and more countries have developed their capabilities in industrial activities, so barriers to entry in production have fallen and the competitive pressures have heightened (Figure 3). This has become particularly apparent since China, with its abundant supplies of educated labour, entered the world market in the mid-1980s.[10] It is this, too, which underlies the falling terms of trade in manufactures of developing countries (see above). Consequently, the primary economic rents in the chain of production are increasingly to be found in areas outside of production. We shall discuss this in more detail in section IV below.

FIGURE 3

COMPETITIVE PRESSURES IN THE VALUE CHAIN



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Governance: A second consideration which helps to transform the value chain from an heuristic to an analytical concept is that the various activities in the chain – within firms and in the division of labour between firms – are subject to what Gereffi has usefully termed 'governance' [*Gereffi, 1994*]. That is, there are key actors in the chain who take responsibility for the inter-firm division of labour, and for the capacities of particular participants to upgrade their activities.

Why is this important? It is because of the nature rather than the extent of trade in the recent era of globalisation. For many countries the trade/GDP shares in the late nineteenth/early twentieth century and the late twentieth century were not dissimilar. The key difference is that in the earlier period this trade was largely in arm's-length relationships, with final products being largely manufactured in a particular country and then exported. By contrast, in the latter period, trade was increasingly in sub-components and services and was consequently considerably more complex [*Feenstra, 1998; Hummels, Ishii and Yi, 1999*]. (This contrast allows us to make the useful distinction between what might be called 'internationalisation' in the late nineteenth century and 'globalisation' in the late twentieth century.) The intricacy and complexity of trade in the globalisation era requires sophisticated forms of coordination, not merely with respect to logistics (who ships what, where and when), but also in relation to the integration of components into the design of the final products and the quality standards with which this integration is achieved. It is this role of coordination, and the complementary role of identifying dynamic rent opportunities and apportioning roles to key players which reflects the act of governance.

This concept of governance – and the distinction between different types of chains – is a major contribution to our understanding of the workings of value chains. What it does is to throw light on those factors determining the nature of the insertion of different producers into the global division of labour. For, as we observed in section I above, it is not just a matter of whether producers participate in the global economy which determines their returns to production, but how and on what terms they do so.

Extending Gereffi's [*1994*] concept, it is possible to distinguish three forms of governance, based on principles of civic governance. The basic rules which define the conditions for participation in the chain need to be set. This can be termed '*legislative governance*'. It is also necessary to audit performance and to check compliance with these rules – this can be seen as '*judicial governance*'. However in order to meet these rules of participation, there needs to be some form of proactive governance (which might be termed '*executive governance*') which provides assistance to value chain participants in meeting these operating rules. Much of the existing discussion of governance fails to recognise this threefold distinction, which

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FIGURE 4

EXAMPLES OF LEGISLATIVE, JUDICIAL AND EXECUTIVE VALUE CHAIN GOVERNANCE

	Exercised by parties internal to chain	Exercised by parties external to chain
Legislative governance	❑ Setting standards for suppliers in relation to on-time deliveries, frequency of deliveries and quality	❑ Environmental standards ❑ Child labour standards
Judicial governance	❑ Monitoring the performance of suppliers in meeting these standards	❑ Monitoring of labour standards by NGOs ❑ Specialised firms monitoring conformance to ISO standards
Executive governance	❑ Supply chain management assisting suppliers to meet these standards ❑ Producer associations assisting members to meet these standards	❑ Specialised service providers ❑ Government industrial policy support

is one of the reasons why there is often confusion about which party actually governs a particular value chain and a reluctance to recognise that different parties may engage in different forms of governance in the same chain. As Figure 4 shows, these governance roles may be provided from within, or from without, the chain.

Building on this concept of governance, Gereffi [*1994*] has made the very useful distinction between two types of value chains. The first describes those chains where the critical governing role is played by a buyer. *Buyer-driven chains* are characteristic of labour-intensive industries (and therefore highly relevant to developing countries) such as footwear, clothing, furniture and toys. The second describes a world where key producers in the chain, generally commanding vital technologies, play the role of coordinating the various links – *producer-driven* chains. Here producers take responsibility for assisting the efficiency of both their suppliers and their customers. In more recent work, Gereffi [*1999a*] has pointed out that producer-driven chains are more likely to be characterised by FDI than are buyer-driven chains. As will be shown in later discussion, governance in individual chains often arises from multiple sources and is frequently contested; moreover, as the sources of rent change, so does the focus of governance activities.

Systemic efficiency: A third analytical element of value chain analysis is that

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it moves the focus of attention from 'point' to 'systemic' (that is, value chain) efficiency. Why might this be important? A useful example can be drawn from the strategic restructuring of one of Europe's major retailers. Tesco has achieved a significant growth in its market share and profitability from its ability to slim-down its own inventories, ensuring a process of just-in-time deliveries from its own warehouses and those of key suppliers to its retail stores. But it has increasingly come to realise that these activities account for only a thin share of the product's total value added and unless it 'governs' its chain to achieve broader levels of systemic integration, little more could be done to achieve competitive advantage. For example, in an exercise designed to identify areas of wasted activity in the value chain of a particular product, an analysis of the value added activities involved in the production and cooling of a tin of coca cola was undertaken. This revealed that in an optimum situation it would take a total minimum time of three hours to produce the complete product. Yet in reality, the actual elapsed time in this process was 319 days, leading to a significant cost in working capital throughout the chain [*Womack and Jones, 1996*]. Tesco has thus begun to put significant resources into trying to improve efficiency throughout the chain, since it has come to realise that the activities which it is directly responsible for in its internal operations account for only a small share of total product costs.

As value chains become increasingly disarticulated and subject to a finer and finer division of labour – an inevitable process given the increasing knowledge content of production – so the ability to make an impact on competitiveness by improving the efficiency of individual links in the chain has become increasingly limited. Systemic integration involves closer cooperation between links in the chain, and this often involves enhanced responsibilities for governors, as well as the growth of greater levels of trust between different links in the chain. Perhaps most importantly from our perspective, value chains increasingly span national boundaries, and governors therefore are also forced to learn how to upgrade producers in low income countries.

How do These Three Analytical Characteristics of Value Chains Relate to our Concerns with Spreading the Gains from Globalisation?

These three analytical elements – dynamic rents, governorship and systemic efficiency gains – are closely linked. The central driver is the prevalence of competition which forces down profits by lowering barriers to entry, and which increases as producers in more and more countries enter global trade.[11] This induces participants throughout the chain to search for new forms of rent. In achieving this, the more powerful actors in the chain are increasingly required to induce (and assist) their suppliers and customers to

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change their own operating procedures. At the same time they continually search for new suppliers (systematically striving to lower barriers to entry in other links in the chain) and customers. These objectives require them to act as value chain governors, although to varying degrees. The resulting growth in the social division of labour requires that these governors act over an increasingly large geographical and institutional terrain to search for systemic efficiency.

We observed in section II that as this global spread of activities has increased, so the inter-country distribution of income has become more complex and generally has worsened. How are these phenomena linked? First, barriers to entry are the determinants of the distribution of rents. That is, they determine who gains and who loses in the chain of production. Those who command rents, and have the ability to create new domains of rent when barriers to entry fall, are the beneficiaries. By contrast, those who are stuck in activities with low barriers to entry lose, and in a world of increasing competition, the extent of these losses will increase over time. Secondly, the growing areas of rent are increasingly found in the intangible parts of the value chain. A distinct development over the past decade has seen the barriers to entry in manufacturing – formerly a key scarce capability lodged predominantly in the industrially advanced countries – begin to fall. More and more countries, with lower and lower wage costs (especially China and India) have developed the capacity to reliably transform physical inputs into high quality physical outputs at a low cost. It is this which explains the diminishing terms of trade of developing countries exports of manufactures observed above. At the same time, copyright and brand-names have a very long life (more than 70 years for the former and in perpetuity for the latter), and these represent 'absolute and immutable' forms of economic rent. It is not surprising, therefore, that the high income countries in general (and the US in particular) have placed so much emphasis on intellectual property rights in recent years. Similarly, as value chains become increasingly complex and subject to coordination, the rent accruing from governorship itself – a particular form of relational rent [*Kaplinsky, 1998*] – has grown.

All of this affects the inter-country distribution of income. But participation in global value chains also affects the intra-country distribution of income. This is because the requirements of final product markets in high income markets invariably require capabilities which are outside of the reach of poor people and poor farms and enterprises (often SMEs). These demands tend to be more exacting in 'governed value chains' than in arm's-length trade. Consequently, small farms and firms and those people reliant for their incomes from smallholdings and small enterprises may suffer, and those locked into large farm production may gain.

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How Does Value Chain Analysis Differ from Conventional Industry Studies and from What Social Scientists (and Especially Economists) Normally do?[12]

Traditionally, the focus on productive activities and the insertion of local producers into global markets has been on the economic branch and the economic sector. Developing countries have been seen to have a potential comparative advantage in the primary branch, and the industrial countries in secondary economic activities and value-added traded services. Within the industrial branch, the focus has most often been on individual sectors (based on ISIC or SITC/HS classifications) such as clothing, shoes, chemicals, electronics, food processing, and in a national context. Hence, the analysis has tended to focus on the size and growth of the sector in terms of employees and gross output (rather than net value added), trade performance and the size distribution of firms.

Value chain analysis throws more light on the determinants of income distribution, both within and between countries, and especially over time than this traditional industry analysis. For example:

- Because it focuses on the dynamics of rent, a value chain perspective forces the analysis to transcend economic branches and sectors. For example, in the forestry and furniture chain, the rent-rich activities are increasingly found in the genetics of seed design and in the design and branding of the furniture, rather than in the individual agricultural, industrial or service sub-sectors (which tend to be the domain of traditional branch and sectoral analyses). It is only through a comprehensive view of the whole chain *that the links in the chain or segments in product markets which are characterised by high or growing rent* can be identified.

- Related to this, value chain analysis makes it possible to trace through a particular thread of *rent-rich activities* which are not easily captured by branch and industry analysis. For example, we have observed that intangible knowledge is increasingly characterised by high barriers to entry, and that the owners of this knowledge gain most from the globalisation of production and exchange. Similarly, in addition to imposing barriers to entry, governance may itself often be subject to significant barriers to entry and hence provide high returns. This being the case, the ability to identify rent-rich activities along the whole chain of added value provides the key to understanding the global appropriation of the returns to production.

- The *data* which are characteristically generated in most branch and sectoral analyses make it difficult to interpret the significance of key

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indicators such as 'output', 'sales' and 'costs'. Consequently the determinants of income distribution are difficult to unravel. Trade statistics are especially problematic here, since they provide little capacity to unpick value added. For example, in the late 1980s, the Dominican Republic saw a significant increase in the gross value of shoe output and exports. But 'shoe production' occurred in EPZs utilising imported inputs – the unit value of a shoe export was a mere $0.23. By contrast unit shoe exports from Italy may more fully reflect value added. In what senses, then, may the shoe sectors in these two countries be compared unless a value chain analysis – incorporating a more sophisticated mapping of input–output relationships – is utilised?

- The dynamic nature of rents generated in the global activities of a value chain are *obscured by a focus on national industries*. For example, when production occurs in the context of falling global product prices, national accounting systems may reflect a growth in activity and value which does not correspond with the international purchasing power of this sectoral activity. The problem is particularly acute when decisions about *national* resource allocation – affecting income streams over time – are made without reference to the *global* dynamics of returns to different activities in the chain. Thus it is the global focus of value chain analysis that more accurately identifies suitable opportunities to augment incomes in a national context than the national focus of industry studies.

- Studies of market structure which fail to locate the analysis within a value chain perspective are not able to adequately explain the *determinants of firm-size distribution*. For example, the high concentration of ownership in the South African furniture industry does not arise from market conduct within the furniture sector. Instead, it is explained by high levels of concentration in the retail sector, which in turn is linked to concentration in financial intermediation [*Kaplinsky and Manning, 1998*]. Similar observations have been made with respect to the footwear industry, but in this case the inter-sectoral linkages which are involved span national boundaries [*Schmitz and Knorringa, 1999*].[13]

So much for the content of research enquiry. But what of the implications for disciplinary focus?

- Because value chain enquiry spans different economic branches and sectors, effective analysis requires the participation of different disciplines. This is most clearly the case in relation to the focus on agricultural and manufacturing production systems, but the focus on the

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dynamics of rent also requires inputs from management studies and engineering. Moreover, since power is a key component of governance, and trust is critical to enhanced inter-firm co-operation and new forms of work-organisation, there is a simultaneous need to draw on the insights of political science and sociology. It is for this reason that Wood reflects that value chain analysis provides 'a *meeting ground* for economics, business administration and industrial sociology in the study of one important aspect of globalisation, namely the simultaneous economic integration of countries, and disintegration of production processes' (emphasis added) [*Wood, 1999: 24*].

- A number of challenges are posed to much of traditional *economic analysis*. The Heckscher-Ohlin factor-price equalisation theorem predicts that in an open economy, factor returns such as wages will tend to converge across (and within some) national boundaries. Yet, this is often not the case, in part due to the falling costs of mobility as highly skilled workers, operating within coordinated value chains, interact with skilled and unskilled workers in different economies [*Wood, 1999*]. The ability to identify and capture the role played by these mobile skills is significantly enhanced when analysis occurs through the lens of the value chain. Much economic analysis of income distribution also tends to focus on the individual as the unit of account, and it is certainly the case that incomes do accrue to individual people as holders of assets (for example, skills and equity). Yet, while individuals may receive incomes, these returns are defined by their participation in institutions (that is, firms) which systematically pursue policies designed to enhance these incomes by constructing barriers to entry against competition. Understanding the processes whereby barriers to entry are constructed takes the analysis beyond the domain of much of economic analysis which treats technological progress as exogenous, and fails to recognise the ability of firms to construct the competitive environment in which they operate (rather than acting as price-takers). Moreover, an understanding of the nature and importance of trust in inter-firm relationships within the value chain requires economists to also engage with the contingency and sociology of the determinants of social capital.
- In a similar way, other disciplines are also forced to rethink their analytical frameworks by a focus on value chains. Wood argues that economics not only provides an accounting framework in which value chains can be mapped, but also forces the enquiry to focus on the economic determinants of location, notably on cost structures [*Wood, 1999*]. Thus, a (complementary) division of labour can be characterised as one in which economists determine the basis of *comparative*

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advantage (that is, the potential which different environments provide for reaping economic rents), and other disciplines identify the determinants of *competitive advantage* (the factors which explain why some firms are able to appropriate these economic rents).

Do these attributes of value chain analysis improve the policy relevance of research? Our primary conclusion is that this analytic framework provides the potential for identifying those policy actions – both by private and public actors – which may influence distributional outcomes. We will return to this in the concluding discussion below, but four observations can be made at this stage:

- As we have seen, value chain analysis covers a range of interconnected economic activities, spanning branches and sectors, and thus provides the potential for coordinating what might be termed '*joined-up policies*' between different arms of government
- By focusing on the institutional determinants of rent, value chain analysis points policy towards instruments which influence the behaviour *of individuals grouped within firms and other organisations* which shape the distribution of returns from production and exchange. Thus, the key to sustaining the growth and spread of incomes lies in the ability to influence the behaviour of groups of people acting in concert, rather than that of individuals who benefit from these joint activities.
- By focusing on the dynamics of rent, value chain analysis points policy towards the *development of those capabilities and institutional trajectories* that can sustain incomes over time.
- Realising systemic value chain efficiency requires inputs from a range of institutions that are not direct participants in the chain themselves, for example producer service providers, scientific institutions and the education sector. This directs the focus of policy attention to the strengthening of the national system of innovation [*Lundvall, 1992*].

These are strong assertions which make powerful claims for the primacy of value chain analysis. But can these assertions be borne out in practice? In the following section, we provide some examples to support these claims, based in large part on the findings of research into a series of selected value chains.

IV. SOME ILLUSTRATIVE CASE STUDIES

In illustrating these claims, we will situate the analysis in terms of the three analytical components of value chain analysis – the changing composition

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of rents, the role played by governors and the systemic nature of chain efficiency – and with a focus on the distributional and policy implications of what has been observed. The case study material considers the production, export and marketing of canned deciduous fruit and of automobile components. Other case studies – of the fresh vegetables and footwear value chains – are provided in other papers in this volume.

As we have seen, the dynamics of economic rent are subject to change as a consequence of increasing competitive pressures which often overwhelm historically significant barriers to entry within different parts of the value chain. This means that static positioning of producers, either within particular activities in particular links of the chain (for example, assembling rather than materials forming), or in particular links in the chain (for example, transforming inputs into outputs, rather than marketing) are likely to be associated with declining terms of trade, and hence with a worsening of relative and/or real incomes.

Table 2 illustrates this dynamic process by summarising the experience of the two value chains which we describe below – canned deciduous fruit (CDF) and automobile components.

Canned Deciduous Fruit[14]

CDF is a processed agricultural product, and by some accounts is thus an ideal sector for low income countries to diversify into and to increase their share of returns in the value chain [*Girvan, 1987*]. Yet the returns accruing to the growers (12.4 per cent) and the retailers in the rich countries (26.7 per cent) are not dissimilar to the case of FFV (Dolan and Humphrey, this collection).[15] In the past, before the EU provided support for European producers, the major sources of rent were to be found in the growing and canning links of the value chain. For this reason, the primary lien on value chain returns accrued to producers in South Africa and Australia, both of which had efficiently organised agro-processing industries. But during the 1980s the European Union began to provide increasing support to the domestic industry, both on the output side (through tariff protection), and via subsidies to fruit inputs. Thus, despite having both lower costs of production and higher standards of quality, southern producers were increasingly squeezed out of global markets by European (particularly Greek) producers. Protected in the European market, they used these trade policy rents to cross-subsidise sales to third markets such as Japan and Latin America. Similarly, US farmers and canners were also protected by a combination of tariffs and in some cases spurious Food and Drug Administration phyto-sanitary standards – for example, rejecting containers of Greek canned peaches on dubious grounds and forcing the exporters to cover the costs of repatriating the contents to Europe.

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TABLE 2

THE DYNAMIC DISTRIBUTION OF RENTS IN TWO VALUE CHAINS

Sector	Links in value chain	Prime source of economic rent			Implications for production activities
		Past	Present	Future	
Canned deciduous fruit	Seed design ↓ Growing ↓ Post-harvest processing ↓ Exporting ↓ Buyers ↓ Retailing	Growers in South Africa and Australia Wholesale	European and US growers and fruit canners Buyers and export agents Brand names	Buyers Retail chain own brands	Growing capabilities (climatic specific) are generalised and competition high; intense competition leads to falling terms of trade. Low barriers to entry mean *that rents are low throughout the chain* *Within individual links*, economic rents are increasingly in intangibles (seed design, growing practices, phyto-sanitary practices, brand names and marketing) Battle between retail chains and brand names for shelf dominance
Automotive Components	Raw material processing ↓ Design ↓ Forming ↓ Assembly ↓ Exporting ↓ OEM user ↓ Spares ←	Design Forming Assembly	Design Coordination of value chain Some in forming and assembly OEM brand name	Design Coordination of value chain OEM brand name	Manufacturing competences become widespread; growth of global sourcing, but intense competition leads to falling terms of trade. Rent achieved *by moving to different links* in the chain. *Within individual links*, economic rents are increasingly in intangibles (design, knowledge inputs into production, brand names and marketing)

* For CDF, see Kaplan and Kaplinsky [*1999*]; for autos and components, see Barnes and Kaplinsky [*2000a, 2000b*], and Humphrey [*2000*].

TABLE 3

BREAKDOWN OF CANNED PEACH VALUE CHAIN

Stage in value chain	Contribution to final product value (%)
Within South Africa:	
Peaches	12.4
Cans	11.6
Sugar	4.2
Canning	14.7
– Labour	7.4
– Other (e.g. depreciation, utilities, profit, internal transport)	7.3
Total inside South Africa	42.9
Outside South Africa:	
Shipping, duties, insurance, landing charges	24.2
Importer's margin	6.3
Supermarket margin	26.7
Total outside of South Africa	57.1

Source: Kaplan and Kaplinsky [*1999*].

Hence, currently, two of the primary recipients of economic rent are the growers and canners in the high income countries. But a second important lien on value chain returns accrued to the holders of brand-named products. The retail chains often sell own brand canned fruit at a loss. They thus pay a premium for major global branded lines (54 per cent), and for other producer branded lines (22 per cent). Faced with this distribution of returns from the value chain, South African producers, attempted to move into these globally branded items, but with limited success. One of the domestic producers bought the Del Monte brand name, but this was only for markets outside of Europe and North America where the brand name carried little weight. In a second case, a royalty was paid to a major brandname holder, but this was so large that it consumed all of the brandname rent generated. (It should be no surprise, but the contents of these brandname products were identical to those of the South African producers' own brands, which sold at a discount of 25 per cent.)

Governance in this chain is relatively simple. 'Legislative governance' is performed by the final product retailers who determine the standards that need to be met, informed by national legislation defining product standards. The auditing of these standards – 'judicial governance' – is performed by a combination of supermarket representatives who visit producers, and the 'category managers', that is the import agents who manage the provision of products to the supermarkets and who, together with the supermarkets search for new sources of supply. 'Executive governance' – assisting

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suppliers to meet the required standards – is provided to the canners by the supermarkets, and to fruit, tin and sugar suppliers by the canners themselves. However, these acts of executive governance are generally modest in nature.

The ability of a national economy based industry to compete is heavily affected by the systemic efficiency of those parts of the value chain which are within the economy. As we can see from the breakdown of value added in South African CDF (Table 3), the canning link's contribution to overall product value added (14.7 per cent) is only just above that added by peaches (12.4 per cent) and the production of cans (11.6 per cent). One route open to the CDF industry, which was experiencing very low levels of profitability in the face of European subsidies, was to enhance the systemic efficiency of the chain. However, low levels of trust and cooperation – endemic to the Apartheid era – meant that achieving these systemic gains was extremely difficult. This surfaced in inefficiencies between all links in the chain – between growers and canners, between steel producers and can-makers, between can-makers and canners, and between canners and sugar manufacturers.[16] These sub-optimal inter-firm links occurred despite attempts being made within each of the links to improve internal efficiency; yet the gains to enhanced inter-firm links far outweighed the benefits accruing to intra-firm improvements. Similarly, there was considerable scope for enhanced inter-firm horizontal linkages, particularly between the canning firms. For example, each firm remarked that if only they presented a united front to the global buyers they would be able to achieve a greater share of value chain returns; instead their rivalry allowed the buyers to play one firm off against another.

There are signs that in the future, rents in this chain are likely to be eroded, with few industry participants sustaining high income levels. Despite the hiccup at the Seattle trade round talks, the trade policy rents which buttress high producer incomes in Europe and North America are almost certainly going to be eroded. Moreover, many locales in the world are suitable for growing deciduous fruit. Marketing rents, too, are under threat with the growth of supermarket concentration in major markets, and the development of supermarket own-brands in a price elastic market. It is significant therefore, that two of South Africa's largest fruit manufacturers are increasingly moving into global sourcing and distribution, and have located offices in Jersey and the UK mainland respectively, at the same time as they are reducing the size of their domestic processing operations.

In terms of spreading the gains from globalisation, future developments are likely to increase the proportion of returns going to low income countries. The major developments in the share of returns over production and processing is likely to be a shift from middle income countries such as

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South Africa to lower income countries such as China. However, over time, as more and more poor economies enter the value chain as producers, these returns are likely to diminish. Moreover, the quality and environmental standards requirements of global buyers are such that within these economies small growers are unlikely to be involved. Moreover, scale-intensive canning (and the production of key inputs such as cans and sugar) are likely to remain in the hands of the large-scale sector. There seems, therefore, little scope for ameliorating adverse distributional trends – within or between countries – in the CDF value chain.

Automobile Components[17]

An automobile is an assembly of more than 5,000 discrete components. Historically, low levels of trust with suppliers involved high transaction costs, so emphasis was given to the internalisation of production by the assemblers who accounted for around 65–70 per cent of the total value of the final product. More recently, following organisational innovations pioneered in Japan which have fostered trust in the supply chain, there has been a tendency to outsource an increasing proportion of components, including the assembly of sub-components. It is not atypical therefore for final assembly plants to now account for 40 per cent or less of final product value.

Various strands of change have come together in recent years to alter the pattern of these sourcing arrangements. First, from being an assembly of heavy metal pieces in the 1950s and 1960s, the automobile has become an increasingly technology-intensive product whose assembly often involves the complex integration of diverse automation technologies. Most assemblers have therefore chosen a path in which they increasingly concentrate on the overall design and systems integration of the product and rely on their core component producers to ensure that the components involve state of the art technologies, both in product and design. A second and related development has been the move towards modular design and sub-assembly in which the first-tier supplier is expected to incorporate a range of components into its sub-assembly, some of which it manufactures and some of which it acquires. Third, the pace of innovation has increased considerably. This requires a parallel arrangement of design and engineering activities, in which assemblers and various tiers of suppliers work simultaneously on the overall product and its components, rather than in the previous sequential and arm's-length pattern. Fourth, new forms of internal production flow and quality assurance – just-in-time production – involve the very close integration of logistics and quality procedures between assemblers and their component suppliers. And, finally, all of this is taking place against a backdrop of the growing liberalisation of trade and

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investment flows in the industry, and with an alarming increase in global overcapacity.

All of these developments in the auto industry have profound implications for the sourcing policies of the assemblers, and indeed for first-tier suppliers from their second-tier suppliers. A key development is the growth of global sourcing. An assembler will work closely with its core suppliers on the design of a new vehicle. This product will, with a few minor modifications, be destined for global or for cross-national regional markets. Once an agreed design of component or sub-assembled components has been reached at source, the assembler will expect to use this design in its global operations. In some cases this may involve an agreed design with two (and occasionally three) component suppliers, but even then it will be on the understanding that these designs will be utilised globally. Thus, for global sourcing to work effectively, it requires a component manufacturer to have an increasingly significant design and technological capability.

But it is not just the design that the assembler has to have confidence in. It also needs to be assured that the components it is acquiring on a global scale will be produced at a competitive price, with adequate quality, conformance to specification. The assemblers, therefore, will commit themselves to *global sourcing* from suppliers, on the understanding that the suppliers will commit themselves in turn to *follower supply*. That is, that they will establish plants located in proximity to all the significant assembly operations of their customers when scales permit. Failing their ability to establish and manage such a diverse group of activities, they will either work in joint ventures with local parties, or license their technology to component producers.

The pattern of governance in this value chain is one in which the major buyers – the assemblers in regard to the first-tier suppliers, or the first-tier suppliers with regard to the second-tier suppliers – set the standards with regard to cost, quality, delivery, participation in e-business buying systems, and so on. Regulatory authorities may also set standards (often pressurised by civil society organisations), for example with regard to recycling and emission standards. This is 'legislative governance'. The same parties who set the standards generally monitor conformance to these standards ('judicial governance'). The ability which individual links in the chain possess to meet these standards is the subject of upgrading inputs by a range of parties ('executive governance'). Critically, the industry now expects that first-tier buyers will undertake responsibility to promote supply chain learning, ensuring that producers in the second-, third- and fourth-tier develop the capabilities to meet the industry's critical success factors [*Bessant et al., 1999*]. But assistance is also provided by specialised service

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providers in this industry, often lodged in the consulting arms of large international accounting firms.

Developments in this global value chain have a number of implications for the inter-country distribution of activities and of returns. On the one hand, the desire to integrate production schedules in assembly with those in component manufacture, and to reduce the costs of holding inventories in the value chain as a whole, has meant that there is considerable scope for decentralised component manufacture. On the other hand, the growing consolidation of the assembly industry, and the desire to standardise production in a limited number of designs (albeit with superficial external modifications) has meant that the global standards of key buyers ('judicial governance') are becoming increasingly important. Thus, component producers need to be connected to final buyers, to be incorporated within a value chain; but in order to do so, they have to utilise global designs. The consequence of global sourcing and follower supply policies by the assemblers and first tier suppliers has meant the erosion of local ownership and local technology in developing countries. Interviews with the assemblers in South Africa show the extent of this problem (Table 4), with a rapid shift by auto assemblers in their sourcing policies as the local industry opened up to global markets. Between 1998 and 2000, the share of wholly-owned subsidiaries amongst the 20 largest South African component suppliers rose from 37 to 60 per cent, and those of wholly-locally-owned firms fell from 58 to 40 per cent; the balance were joint ventures [*Barnes, 2000*]. In Brazil, in 1995, of the largest 25 Brazilian component manufacturers 12 were wholly or majority, local-owned. By the end of 1998, five of these had been taken over by foreign firms, and only one out of the 13 largest suppliers remained in domestic ownership [*Humphrey, 2000*].

TABLE 4

COMPONENT SOURCING BY FOUR AUTO ASSEMBLERS IN SOUTH AFRICA, 1997, 2000 AND 2003 (PROJECTED)

Components sourced from	1997	Presently	2003
Wholly owned subsidiaries of MNC auto component manufacturers	26%	31.7%	37.5%
Joint ventures between SA companies and MNC auto component manufacturers	18.5%	26%	32.5%
SA companies with technology agreements with MNC auto component manufacturers	29.8%	24.3%	20%
SA companies with SA technologies	25.8%	18%	10%
Total	100%	100%	100%

Source: Barnes [*2000*].

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Where does this leave low income country producers of automobile components? In terms of the inter-country distribution of returns, it means that the design activities in this value chain – where the rents are increasingly to be found – lie in the hands of a decreasing number of global components manufacturers. Their core design activities are being undertaken in design centres in high income countries, with only peripheral design activities occurring in low income countries. From the perspective of independent low income country component suppliers, there are few alternatives to this global division of labour. They can either manufacture to the designs of the global first- and second-tier component suppliers, or produce spare parts for the aftermarket. But this, too, is heavily dominated by global brand names (where the rent is concentrated). The unbranded aftermarket is highly competitive, often exhibiting low barriers to entry and is subject to growing competition in production as buyers systematically cultivate competition from new producers. In regard to the intra-country distribution of returns, this is a sector with decreasing space for SMEs (insofar as they may be a conduit for spreading the gains from participation in global markets), so that the key instrument of distribution is likely to lie in wage rates. Here, on the one hand, wages in this sector are generally relatively high in most countries by comparison with other sectors; on the other hand, the component sector is being subject to increasing competition,[18] and it is possible that the privileged status of these workers will be eroded over time.

V. CONCLUSIONS

The objective of this study is to illustrate how value chain analysis can contribute to a better understanding of the determinants of inter- and intra-country income distribution, and consequently also to the identification of policies which might improve the distributional outcome of countries' insertion into the global economy. We have argued that a focus on three elements of value chains – the dynamics of rents within the chain, the governance of the chains and their transnational systemic character – is necessary to turn an heuristic construct into an analytical tool.

In the preceding section we presented two case studies illustrating the way in which value chains – understood in their analytical sense – can explain why some parties gained, and some lost from the globalisation of production and exchange. They consequently were able to contribute to an explanation, rooted in production and exchange (rather than in finance or other spheres of globalisation), of why the growing global spread of economic activity has not been associated with an equivalent spreading of the gains from this economic activity.

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But can value chain analysis tell us about the policies which are required to reverse these malign outcomes? Here, two key insights are provided by value chain analysis. First, global production networks are becoming increasingly complex and arm's-length trade is increasingly being confined to commodities with low returns. Access to high-income yielding activities therefore requires participating in global value chains, and the key challenges are thus to identify ways in which poor countries and poor producers can both enter these chains, and participate in ways which lead to sustainable income growth. Second, the analysis of governance relations which permeate value chains leads to the identification of the major institutional actors, and this provides insights into policy levers which might influence the behaviour of key stakeholders in the value chain.

Value chains are not homogeneous, and the opportunities for rent appropriation by different parties will vary. Essentially, however, when threatened by competition, there are four directions in which *economic actors* can move; these paths are not mutually exclusive:

- increasing the efficiency of internal operations such that these are significantly better than those of rivals;
- inter-firm linkages can be enhanced to a greater degree than that achieved by competitors;
- introducing new products or improving old products faster than rivals;
- changing the mix of activities conducted within the firm or moving the locus of activities to different links in the chain, for example from manufacturing to design.

The analysis of the two value chains in section IV above suggests that on their own, the first two steps are unlikely to realise a greater share of value chain returns. This is because these capabilities are increasingly widely diffused through the global economy (underlying the falling terms of trade of developing countries trade in manufactures observed in section II) and concentration in these areas (particularly in intra-firm efficiency) may well lead to immiserising growth. In other words, they are generally only the necessary conditions for an enhanced spreading of the gains from participating in global markets. It is the third and fourth steps that are likely to provide the greatest results. But here, poor producers and producers in poor countries run up against relations of power, embedded in value chain governance (Figure 5). These are highly protected domains, precisely because they are repositories of economic rent. As Schmitz and Knorringa (this collection) observe with regard to the footwear value chain, the global buyers who dominate this chain are happy to assist developing country producers in the first three of these policy alternatives, but zealously exclude them from the fourth.

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FIGURE 5

SCOPE FOR ACTIONS TO PROMOTE THE SPREADING OF GAINS TO LOW INCOME PRODUCERS

	Upgrade within link	Upgrade relationship between firms	Redefining activities undertaken within links	Move to new links in the chain
Canned deciduous fruit	Special assistance needed for small growers	Need to improve vertical and horizontal cooperation, especially in relation to participation by SMEs	Little scope for additional action	May be possible to build global or regional brand names
Automobile components	Special assistance needed for SMEs	Supply chain efficiency and learning are critical, especially in relation to participation by SMEs	Possible for first-tier suppliers mainly, but few in low income countries	Difficult as buyers block move into design

The fact that the move into other links in the chain is difficult does not reduce the necessity for action, since in almost all value chains this is clearly the optimal route. It also does not mean that firms should always pursue this path at the expense of upgrading their internal operations, in improving their links with other parts of the domestic chain or in increasing the range of activities or repositioning themselves within the link in which they already operate. In each of the four cases, there is scope for pursuing multiple objectives.

It is likely that the same conclusion will apply to many other value chains in which poor countries operate. But in some cases, the judgement may be that barriers to entry are so low throughout the chain that it will never support acceptable levels of income. This, for example, may be the case in CDF, where producer surpluses are systematically channeled into consumer surpluses through the competitive process and where, over time, the returns to production are whittled away in 'a race to the bottom'.

So much for the policy response from the corporate sector. But, *what can governments do*? Governments have a number of key roles to play. First, they can proactively assist the private sector, workers' organisations and other stakeholders to recognise the opportunities and threats posed by

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participating in global value chains. Secondly, measures can be taken to assist producers to enter these chains. Thirdly, various policy instruments can be used to support the repositioning of the corporate sector within value chains so that they can derive a greater share of the gains. (This occurred in the UK through the Enterprise Initiative Programme during the 1980s, for example). Support can be provided for the enhancement of design skills, as the Spanish government showed with regard to the clothing and footwear industries in the same decade, and as local government does in Brazil for footwear [*Schmitz, 1999*].

But, fourthly, producer rents are not the only form of economic rent which may bolster the returns accruing to poor countries and poor producers [*Kaplinsky, 1998*]. Producers require relatively good access to a range of complementary assets. For example, as Wood and Jordan show for Uganda, physical infrastructure is an important determinant of participating in global product markets, particularly in the manufacturing sector (Wood and Jordan, this collection). In the past this meant relatively good roads, railways and ports but increasingly it also applies to telecommunications. Other forms of rent where government has a role to play include financial intermediation (relatively smooth access by producers to adequate levels of productive capital, which may or may not be provided by 'efficiently functioning financial markets') and human resource rents (access to relatively skilled workers at relatively low wages). Critically, it also includes trade policy rents, where the government either negotiates trade policy privileges, or acts to erode the privileges open to other producers. It is also important that governments help their producers – especially poor producers – to take advantage of trade policy rents, since in many cases low income countries do not fill their quotas or benefit from preferential tariffs in high income markets.

Value chain analysis is crucial to this joined-up policy support because it enables governments to focus on the dynamics of rent, on the pervasive and complex nature of support which is required to build institutions and on managing the integration of individual sectors (and of the whole economy) into the global economy in a manenr which provides for sustainable and equitable income growth.

In the preceding discussion on individual value chains (section IV above), the focus was predominantly placed on the insights provided by value chain analysis to the determinants of the inter-country distribution of income. Yet, increasing inequalities are not just being experienced at the inter-country level, but also with regard to intranational distribution, affecting different regions, different sized firms, different households and different genders. But here, too, value chain analysis has a role to play, particularly in regard to those production structures which involve

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international exchange. What it does do is to illustrate how critical success factors in external markets, allied to governance structures within individual chains circumscribe the role played by domestic actors. We have seen the implications which this has for SMEs in the South African furniture industry (and for small farms in fresh fruit and vegetables – Dolan *et al.*, this collection), but this is an underexplored issue, particularly with regard to gender and household distributional factors.

In summary, it is not so much that value chain analysis tells us anything new, for most of these policy responses have found their way into corporate and government decision-making in many chains and in many sectors. But what it does do is to provide a comprehensive framework for a 'joined-up' series of responses by a range of stakeholders which forces us into a wider, dynamic and more strategic consideration of these issues. Partial analysis and partial responses are likely to be severely sub-optimal in meeting the challenges which are posed, particularly in relation to medium and long run positioning of poor producers and poor countries. To return to the beginning of this paper, positioning and path dependency are critical since participation in the global economy in itself may not provide a path to sustainable income growth or to an equitable distribution of returns.

NOTES

1. For data on the simultaneous advance of global integration of production and exchange, as well as various indicators of inequality, see Kaplinsky [*2000a*].
2. The falling manufacturing terms of trade of developing countries were hypothesised in the early 1970s by Hans Singer [*1971*]. A recent study of the barter terms of trade in manufactures between developing countries and the European Union estimates an annual rate of depreciation of 2.2 per cent between 1979 and 1994 [*Maizels et al., 1998*]. A further study, on the terms of trade in manufactures between the US and developing countries for the period 1981–97, conclude that '[o]ver the whole period, the relative terms of trade trend of developing countries, compared with that of developed countries, has significantly worsened' [*Maizels et al., 1999: 23*]. It is significant that neither of these recent studies reflect the fall in developing country manufactured export prices following the East Asian crisis of 1997–98.
3. For data on China's rapidly growing share of global markets in labour-intensive products, see Kaplinsky [*2000b*].
4. The ability of Dominican Republic firms to upgrade was constrained by the terms of the Caribbean Trade Initiative which limited their capacity to utilise local textile inputs [*Kaplinsky, 1993*].
5. An extended discussion of immiserising growth needs to take account of the purchasing power parity of incomes, whether falling barter terms of trade are associated with falling income terms of trade, and of the opportunity costs of exporting activities in the context of falling barter terms of trade. Some of these issues are discussed further in Kaplinsky and Readman [*2000*].
6. Literally, the word *filière* means 'thread', and was first used in the 1960s in the analysis of agricultural policy in the French colonies and then in late 1970s and early 1980s to describe the perceived need for French industrial capability to span the complete thread of a value chain (Kydd, Pearse and Stockbridge, 1996, cited in Raikes, Friis-Jensen and Ponte [*2000*]).
7. Unfortunately, the phrase 'value chain' covers both the heuristic and analytical categories.

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This has led some to search for a different nomenclature. For example, Gereffi [*1994*] coined the phrase 'global commodity chain' (GCC) and in a recent contribution argues that the GCC is distinct in that it incorporates an international dimension, that it focuses on power of lead firms and the coordination of global activities, and that it explicitly recognises the importance of organisational learning [*Gereffi, 1999b*]. These are proximate to the three characteristics addressed in this paper. But, although representing a major contribution to our thinking on global production networks, the *phrase* 'global commodity chain' suffers because the word 'commodity' implies the production of undifferentiated products in processes with low barriers to entry. The problem with this, as we shall see below, is that the search for sustainable income growth requires producers to position themselves precisely in non-commodity, high barriers to entry activities in the value chain. For these reasons, and in the absence of an agreed phraseology, we will continue to use the words 'value chain', but to do so in an analytical context.

8. For a longer discussion of economic rent see Kaplinsky [*1998*].
9. Although monopoly rent (as defined by the rent-seeking literature) also arises as a result of purposive action, it is usefully distinguished from various forms of innovation rents that reflect the search for new combinations' in the pursuit of entrepreneurial surplus.
10. The share of manufacturing in Chinese exports rose from 49.4 in 1985 to 85.6 per cent in 1995 [*Khan, 1999*].
11. I am grateful to Hubert Schmitz for pointing out this quote: '[t]he GCC approach explains the distribution of wealth within a chain as an outcome of the relative intensity of competition within different nodes' [*Gereffi, Korzeniewicz and Korzeniewicz, 1994: 4*].
12. This discussion has been helpfully informed by memos prepared for the Spreading the Gains from Globalisation research network by Gereffi, Humphrey and Sturgeon [*2000*] and Wood [*1999*].
13. As a consequence, the growing concentration in the Italian, Spanish and Greek retail sectors is likely to undermine the historic strength of the SME sectors in these countries.
14. This discussion is drawn from Kaplan and Kaplinsky [*1999*].
15. These returns are to supermarket own brands. As we shall see later, branded products sell at a considerable premium. In the case of these branded goods, the share of returns going to the fruit growers and processors are significantly lower.
16. See Kaplan and Kaplinsky [*1999: 1795*] for details of these inter-linkage weaknesses.
17. This discussion is based on Humphrey [*2000*] and Barnes and Kaplinsky [*2000a, 2000b*].
18. This is a sector subject to systematic 'cost-down' (that is, price reduction) pressures. Between 1995 and 2000 unit component prices in the USA fell by 3.5 per cent p.a. (in nominal terms, and greater in real terms given price inflation)

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